

## M23DE0207 Java Lab

### Part A

(Minimum 80% of programs are mandatory)

| Sl. No. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Hrs | COs |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|
| 1       | <p>Demonstration of Programs on Command Line Arguments</p> <pre>import java.util.*; class Largest {     public static void main(String args[])     {         int n1, n2, n3, big;         n1=Integer.parseInt(args[0]);         n2= Integer.parseInt(args[1]);         n3= Integer.parseInt(args[2]);         if(n1&gt;n2 &amp;&amp; n1&gt;n3)             big=n1;         else if(n2&gt;n1 &amp;&amp; n2&gt;n3)             big=n2;         else             big=n3;         System.out.println("Biggest No: " + big);     } }</pre> <p>Output:<br/>C:\Users\reva00381\Desktop\Java&gt;javac Largest.java<br/>C:\Users\reva00381\Desktop\Java&gt;java Largest 10 34 54<br/>Biggest No: 54</p> |     |     |
| 2       | <p>Demonstration of Creation of Class, Objects</p> <pre>import java.util.*; class Factorial {     int n;     void fact()     {         int i, fact=1;         Scanner s= new Scanner(System.in);         System.out.print("Please Enter a No.");         n=s.nextInt();         for(i=1;i&lt;=n;i++)             fact=fact*i ;         System.out.println("Factorial of " + n + " is " + fact);     }     public static void main(String args[])     {         Factorial f=new Factorial();     } }</pre>                                                                                                                                                                                      |     |     |

```
f.fact();
}
}
C:\Users\reva00381\Desktop\Java\Java>javac Factorial.java
C:\Users\reva00381\Desktop\Java\Java>java Factorial
Please Enter a No.4
Factorial of 4 is 24
```

```
C:\Users\reva00381\Desktop\Java\Java>java Factorial
Please Enter a No.8
Factorial of 8 is 40320
```

### 3 Programs on Method Overloading and Constructor Overloading

#### i. Method Overloading

class overload\_method

```
{
void add(int a, int b)
{
System.out.println("sum = " +(a+b));
}
void add(String s1, String s2)
{
System.out.println("Concat = " +(s1+s2));
}
public static void main(String[] args)
{
overload_method o=new overload_method();
o.add(23,12);
o.add("REVA","University");
}
}
```

Output:

```
C:\Users\reva00381\Desktop\Java>java overload_method
sum = 35
Concat = REVAUniversity
```

#### ii. Constructor Overloading

class Rectangle

```
{
int length;
int width;

void area()
{
System.out.println("Area=" +(length*width));
}
}
```

Rectangle()

```
{
length=10;
width=10;
```

```

    }
    Rectangle(int a)
    {
        length=width=a;
    }
    Rectangle(int a, int b)
    {
        length=a;
        width=b;
    }
}
class const_over
{
public static void main(String args[])
{
    Rectangle r1= new Rectangle();
    Rectangle r2=new Rectangle(5);
    Rectangle r3=new Rectangle(20,10);
    System.out.println(" ");
    r1.area();
    r2.area();
    r3.area();
}
}

```

Output:

C:\Users\reva00381\Desktop\Java>java const\_over

Area=100

Area=25

Area=200

#### 4 Programs on Inheritance and Method Overriding

i. Simple Inheritance with method Overriding

```

class Room{
    int length;
    int breadth;
    Room(int x, int y)
    {
        length=x;
        breadth=y;
    }
    void area()
    {
        System.out.println("Baseclass_Area" +(length*breadth));
    }
}
class StudyRoom extends Room{
    int height;
    StudyRoom(int x, int y, int z)

```

```

{
super(x,y);
height=z;
}
void area()
{
super.area();
System.out.println("Derived class_Area"+(length*breadth*height));
}
}
class MethodOverride{
public static void main(String args[]){
StudyRoom sr = new StudyRoom(14,12,10);
sr.area();
}
}

```

## ii. Multilevel Inheritance

```

class Shape {
    public void display()
    {
        System.out.println("Inside display");
    }
}

class Rectangle extends Shape {
    public void area()
    {
        System.out.println("Inside area");
    }
}

class Cube extends Rectangle {
    public void volume()
    {
        System.out.println("Inside volume");
    }
}

public class multilevel {
    public static void main(String[] arguments) {
        Cube cube = new Cube();
        cube.display();
        cube.area();
        cube.volume();
    }
}

```

Output:

```
C:\Users\reva00381\Desktop\Java>java multilevel
Inside display
Inside area
Inside volume
```